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Late Cainozoic rotation of the Peruvian Western Cordillera and the uplift of the Central Andes

Orlando Macedo-Sánchez [a,b], Jérôme Surmont [a], Catherine Kissel [a], Pierre Mitouard [a] and Carlo Laj [a]

[a] *Centre des Faibles Radioactivités, C.N.R.S.–C.E.A., Avenue de la Terrasse, 91198 Gif-sur-Yvette Cedex, France*
[b] *Instituto Geofisico del Peru, Casilla 3747, Lima 100, Peru*

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ABSTRACT

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A palaeomagnetic study was carried out in the region of Lima (Peru), on 55 sites (564 cores) of middle Cretaceous to Late Miocene age, distributed along three northeast–southwest transverses which cross the Western Cordillera of the Central Andes. Over the entire studied area, the results consistently yield westerly declinations corresponding to post-Cretaceous and post-Eocene counterclockwise rotations with respect to stable South America, of $16° \pm 6°$ and $13.7° \pm 8.1°$, respectively, when compared to the ill-defined reference poles of this continent. These new results are in good agreement with previous ones from the Central Peruvian Andes and support the model recently proposed for the uplift of the Andean Cordillera.

Introduction

The Arica deflection, for which the concept of oroclinal bending was originally proposed by Carey (1955), is one of the major structural features of the Central Andes. It is defined by a change in the strike of the major structures from a northwest–southeast to a north–south direction, at about 19°S. In this region, the crustal thickness and the mean elevation of the mountain range increase and are accompanied by the development of the Altiplano (James, 1971; Wilson, 1985). These characteristics led Isacks (1988) to propose a model relating the uplift of the Central Andean Plateau to the bending of the Bolivian Orocline. In this model, an enhancement of the concave shape of the Arica elbow originates from along-strike variations in the amount of late Cainozoic shortening. These, in turn, arise from corresponding variations of the width of the weakened zone in the overriding plate (Kono et al., 1985, 1989; Fukao et al., 1989). This model is in good agreement with previous Cretaceous and Palaeogene palaeomagnetic data, which indicate counterclockwise rotations north of the deflection, clockwise south of it, and negligible rotations in the region of the deflection itself (Palmer et al., 1980; Heki et al., 1983, 1984, 1985a,b; Turner et al., 1984; Kono et al., 1985; McFadden et al., 1985, 1990; May and Butler, 1985; Beck et al., 1986; Tanaka et al., 1988; Hartley et al., 1988; Mitouard et al., 1990).

In contrast with Isacks' model, Beck (1987, 1988) has proposed that, similar to the North-American Cordillera, the palaeomagnetic results reflect in-situ block rotations in response to the oblique convergence of the Nazca plate towards

Correspondence to: O. Macedo-Sánchez, Centre des Faibles Radioactivités, C.N.R.S. - C.E.A., Avenue de la Terrasse, 91198 Gif-sur-Yvette Cedex, France.

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the South-American coast. In this type of model, the blocks, which rotate individually around vertical axes, are localized in a zone of distributed shear characterized by major strike-slip faults parallel to the coast. The blocks themselves are limited by many strike-slip faults oblique with respect to the previous ones (Beck, 1980). It has been suggested that this model could account for the rotations which postdate the accretion of allochthonous blocks in southern Ecuador and northern Peru where such faults have been recognized (Mégard et al., 1987; Roperch et al., 1987; Mourier et al., 1988; Laj et al., 1989; Mitouard et al., 1990). In the Central Andes of Peru, on the contrary, one may question whether Beck's model provides a realistic explanation for their tectonic evolution, because no major strike-slip faults appear to disturb the regularity of the Cainozoic folds. In only two regions, the Cajamarca area in the northern Central Andes and the Cuzco area, the folded Pre-Cainozoic sedimentary formations are affected by minor deflections caused by the sinistral reactivation of east–west palaeofaults (Mégard, 1987; Mourier, 1988); but this limited amount of faulting does not appear to have the amplitude and the widespread character required for block rotations to be a reasonable explanation of the rotated palaeomagnetic directions.

In order to extend the results to regions of the Peruvian margin characterized by structures of purely Andean trend, we have carried out a palaeomagnetic sampling in the region of Lima. Fifty-five sites distributed along three northeast–southwest transverses crossing the whole Western Cordillera up to the Altiplano (Huacho–Oyon, Lima–La Oroya and Canete valley) were sampled and studied in the laboratory. These sites correspond to sedimentary, volcanic and intrusive formations of middle Cretaceous to Late Miocene ages. The obtained results were interpreted in terms of rotations with respect to Stable South America (SOAM) after discussion of the available palaeomagnetic reference poles.

Geology

At the latitude of Lima, the Western Cordillera stretches from the coast to the Altiplano. In its western part, it comprises outcrops of late Jurassic to middle Cretaceous sedimentary formations, covered during Albian times by the pillow lavas, tuffs, volcaniclastic sediments and black marly-limestones of the thick Casma Group (Cobbing, 1985). These formations were intruded between 82 and 34 Ma by several tonalitic and granodioritic massifs (Pitcher, 1985; Pitcher and Cobbing, 1985) which constitute the coastal batholith, and were covered by the Eocene to Miocene lava flows and tuffs of the Calipuy Group. In its eastern part, the Western Cordillera comprises essentially strongly folded Mesozoic sedimentary formations including the late Aptian to Turonian Pariahuanca, Chulec, Pariatambo and Jumasha formations (Jaillard, 1987; Macellari, 1988).

The Andean deformations in the Western Cordillera are characterized by thrusts and folds with an eastward vergence. They started during Late Cretaceous with the Santonian Peruvian phase (Mégard, 1984), then were followed by the Mid- to Late Eocene Incaic phase (~ 40 Ma) responsible for the major shortenings (Mégard, 1987), and finally, by the three Miocene Quechua phases (~ 20, 10 and 5 Ma, respectively, Noble et al., 1974; McKee and Noble, 1982; Mégard et al., 1984). Continental red molasses, volcanic and volcano-sedimentary deposits constitute the post-tectonic formations of these different phases.

Sampling

The location of the studied sites is shown in Figure 1. Samples were collected in the Cretaceous grey or dark-grey limestones of the Pariahuanca, Chulec, Pariatambo and Jumasha formations (18 sites, 207 cores), in the Albian pillow lavas and volcaniclastic series of the Casma Group (12 sites, 119 cores), in the Eocene to Miocene lavas and tuffs of the Calipuy Group (4 sites, 41 cores), and in the Cretaceous to Cainozoic intrusive and effusive formations of the coastal batholith (21 sites, 197 cores). The cores were drilled using a portable drill, and oriented in situ using both a magnetic and a sun compass in order to correct for possible local magnetic disturbances. The cores were subsequently cut in several cylindrical samples of standard dimension.

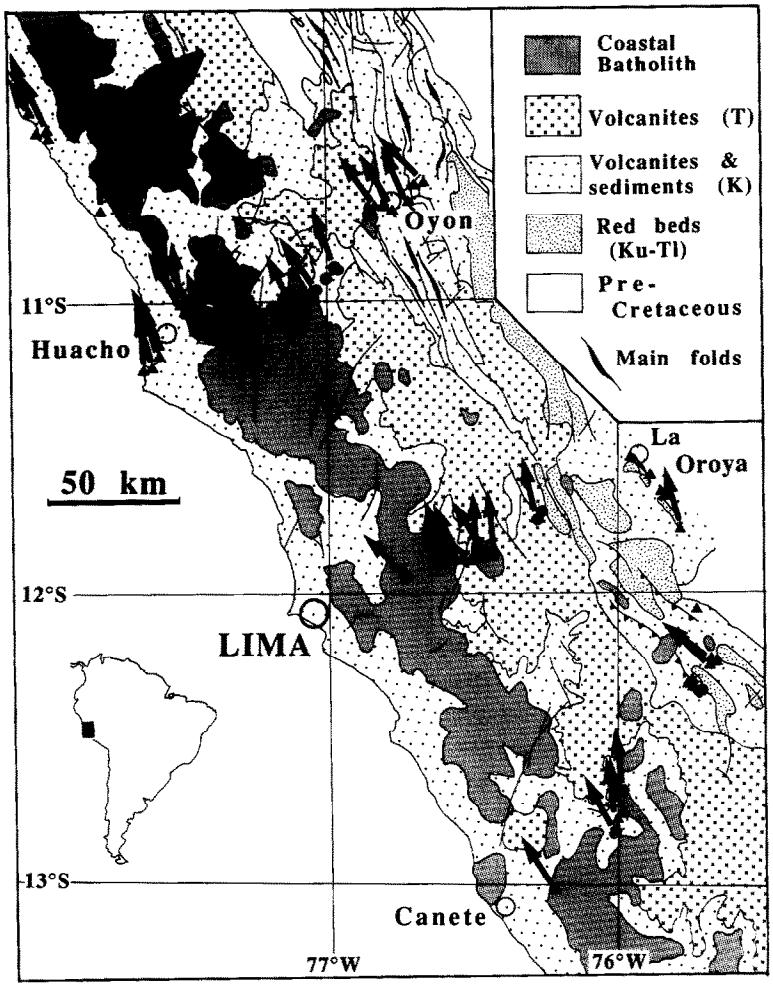


Fig. 1. Simplified geological map of the Western Cordillera in the region of Lima. Triangles and circles correspond to the locations of Cretaceous and Cainozoic sites, respectively. The arrows indicate the mean declinations obtained for each reliable site.

Clear bedding plane attitudes were always obtained for the sedimentary formations. For the volcanic and intrusive series, only slightly deformed since Oligocene times in this part of the cordillera (Noble et al., 1974), a sedimentary control was sometimes available and always yielded minor tilt corrections. No correction was applied when no sedimentary control was found in the field.

Palaeomagnetic analysis

Magnetic mineralogy

Several pilot-samples from the different sampled formations were submitted to magnetic mineralogy analyses. For the volcanic or intrusive samples, these analyses were made using magnetic extracts obtained with a method similar to the one described by Papamarinopoulos et al. (1982). Thermomagnetic curves (during Curie-balance analyses) are generally reversible and most of them indicate a Curie point at 580°C which characterizes magnetite (Fig. 2).

For the sedimentary samples, the small quantity of available material and its very low content in magnetic minerals did not allow extraction and Curie-balance analysis. In this case, the magnetic minerals were identified by investigating the stepwise acquisition of the isothermal remanent magnetization (IRM) up to 1.5 T, and the thermal demagnetization curves of the saturation IRM

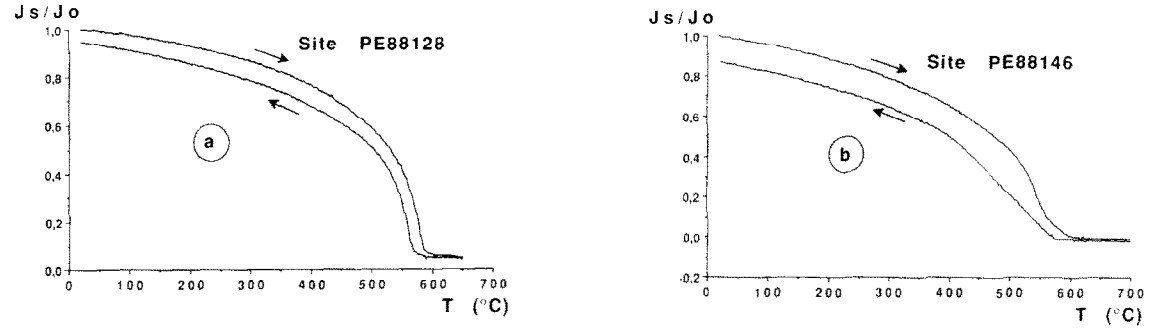


Fig. 2. Thermomagnetic curves obtained by Curie-balance analysis, for a Late Cretaceous intrusive rock sample (a) and an Albian volcanic rock sample (b) from the Casma Group (see text).

(SIRM). Measurements of the IRM were made with a DIGICO spinner magnetometer. After reaching saturation, the samples were thermally demagnetized at steps of 25°C, up to complete removal of the magnetization. After each heating step, the low field magnetic susceptibility of the samples was measured with a DIGICO bridge, in order to reveal any mineralogical transformation occurring during the heating.

Some of the samples (PE136, Fig. 3a,b) reached complete saturation at 0.2–0.3 T, and during their subsequent heating the intensity of the SIRM decreased regularly and was removed at about 350°C. Such behaviour characterizes a simple

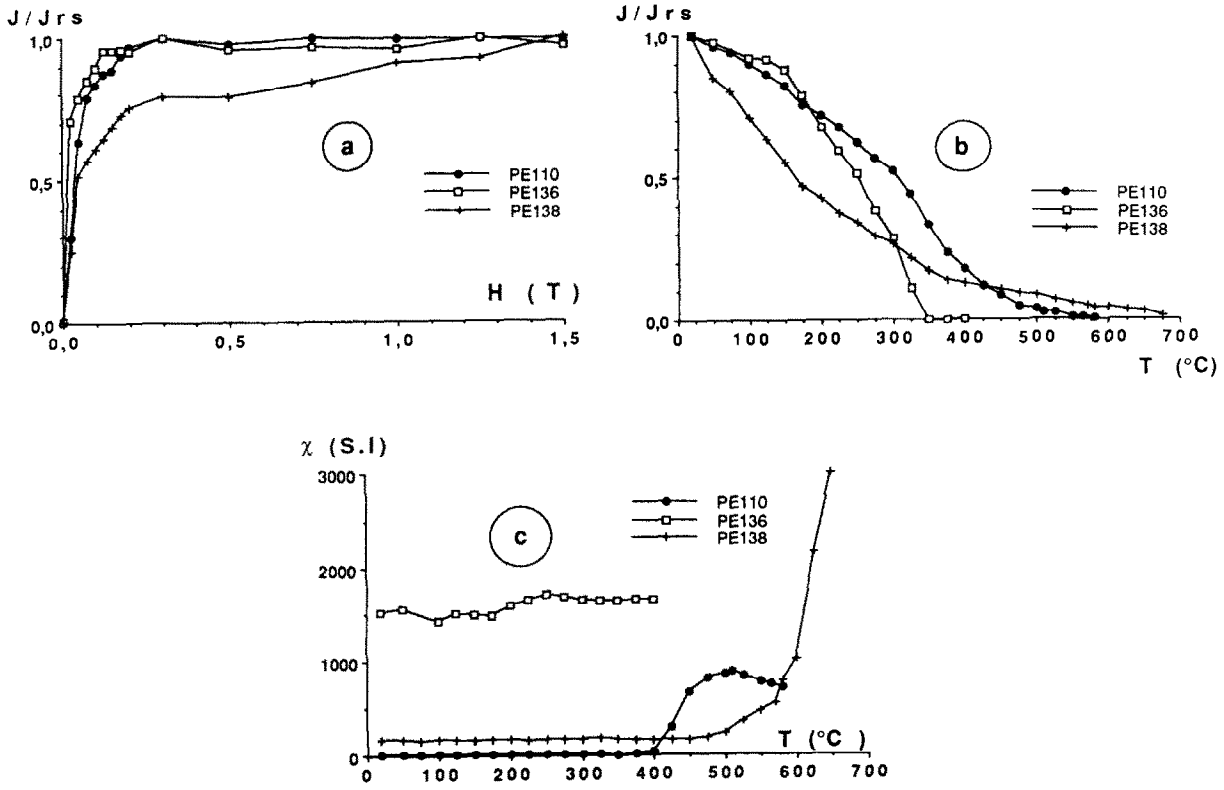


Fig. 3. Normalized IRM curves (a), and subsequent thermal demagnetization curves of the maximum IRM (b) for three samples from the Albian Pariatambo (PE110 and PE136) and Pariahuanca (PE138) sedimentary formations. (c) Low field magnetic susceptibility measurements versus temperature for the same samples. Explanations in the text.

magnetic mineralogy with a single magnetic mineral, most likely an iron-sulphide of pyrrhotite or greigite type.

Other samples (PE110; Fig. 3a,b) are also characterized by a rapid saturation, but during the subsequent thermal demagnetization, the intensity of the SIRM decreases gradually up to 580°C with a slight change in the slope of the curve around 400°C. Most probably, these samples contain magnetite in association with iron-sulphides (greigite?) which themselves turn into magnetite after 400°C, as suggested by the abrupt increase of the magnetic susceptibility beyond this temperature (Fig. 3c).

Finally, the rest of the samples (PE138, Fig. 3a,b) show complex magnetic mineralogy which is characterized by an initially rapid and then slower increase of their IRM, which does not reach saturation at 1.5 T. This indicates, in addition to magnetite, the presence of haematite (confirmed by final demagnetization near 700°C). Part of this haematite might have originated from the transformation of non-magnetic minerals during the heating after 500°C as indicated by the strong increase of the susceptibility.

Palaeomagnetic directions

Measurements were made with a 2G 3-axis-755R cryogenic magnetometer. All the samples were thermally demagnetized in 12 to 15 steps up to the complete elimination of their remanence, by progressive heatings performed between 20°C and 700°C maximum.

The natural remanent magnetizations (NRM) of samples from the volcanic or intrusive series are always high, of the order of 25×10^{-2} A/m. Generally these samples are characterized by a single stable component of magnetization which is removed beyond 250–300°C up to the end of the treatment (Fig. 4a). Occasionally, the thermal demagnetization was not successful and led to

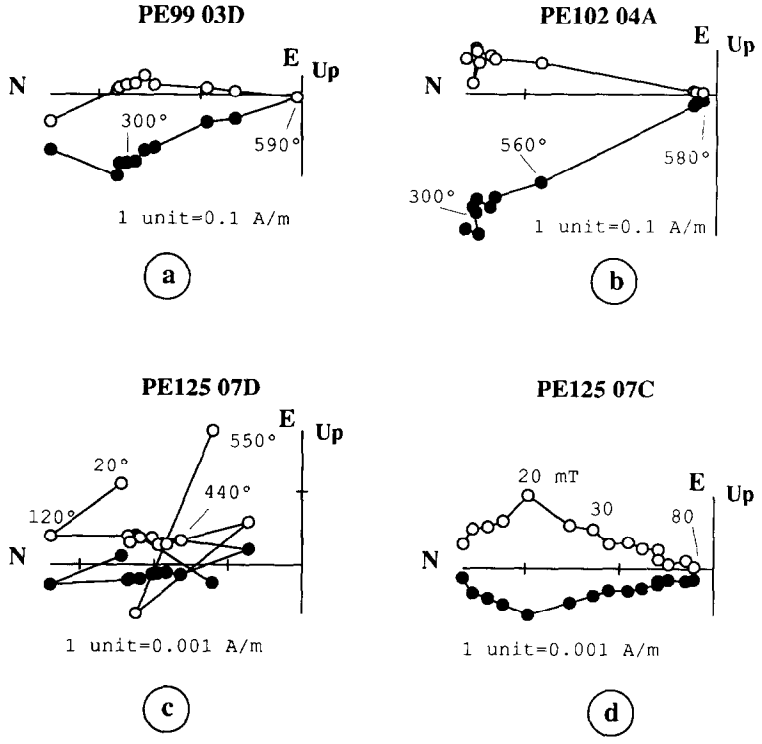


Fig. 4. Orthogonal projection diagrams of the thermal demagnetization of a Late Cretaceous intrusive rock sample (a) and of an Albian volcanic rock sample (b) from the Casma Group, with a single stable component of magnetization. (c) Unsuccessful thermal demagnetization of a volcanic rock sample from the Casma Group due to mineralogic transformations occurring during heating. The diagram is shown after tilt correction. (d) Successful alternating fields demagnetization of the same sample. Full and open symbols represent projections on the horizontal plane and the north–south vertical one, respectively.

incomplete results with incoherent directions beyond 450°C, probably because of mineralogical transformations occurring during heating. The alternating fields treatment was tested on these samples, and yielded in some cases a stable palaeomagnetic direction (Fig. 4c).

The NRMs of samples from the Cretaceous sedimentary series are much lower, in the order of 7×10^{-4} A/m. A single stable component of magnetization was sometimes observed in these samples, but more frequently at least two components were revealed with widely different directions acquired at different moments of the history of the rock. In only 40% of the sedimentary sites studied, the unstable components were easily removed and a stable final direction could be isolated (Fig. 5a). For the other sites, whatever the demagnetization technique, the different components, carried by different families of magnetic particles as it was revealed in the mineralogy analyses, remain unresolved to the very end of the demagnetization (Fig. 5b). In these cases, the direction of the remaining magnetization follows a path along a great circle of projection, i.e. a remagnetization circle (Halls, 1978; Kirschvink, 1980; Surmont, 1989), but without reaching a stable endpoint (Fig. 5c). The large amplitude of some of these paths suggests that these samples comprise sub-antiparallel components of magnetization, the directions of which cannot be precisely determined.

Of the 55 sites sampled in the whole Western Cordillera in the vicinity of Lima, 39 gave reliable results, i.e. a stable final component. The charac-

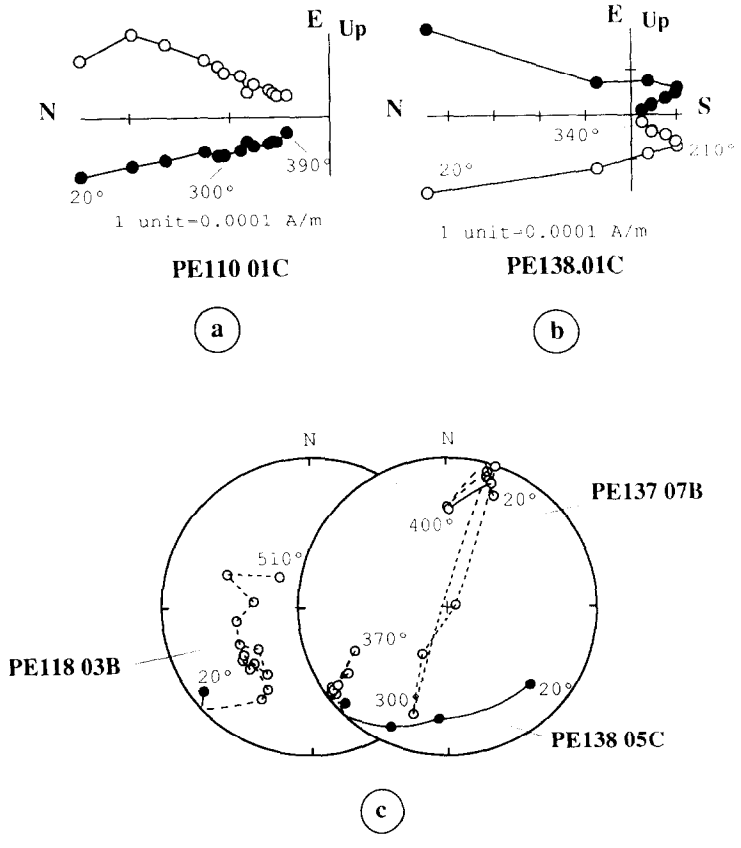


Fig. 5. Examples of different behaviours of samples from the Albian Pariatambo (PE110, PE118, PE137) and Pariahuanca (PE138) sedimentary formations during thermal demagnetization. Orthogonal projection diagrams for a sample with a single stable component of magnetization (a), and for a sample with two components partially superimposed (b). Same conventions as Fig. 4. (c) Stereoplot of three different vector paths of the total remanent magnetization during thermal cleaning corresponding to three different samples with two superimposed components of magnetization. Full and open symbols represent downward and upward vectors, respectively.

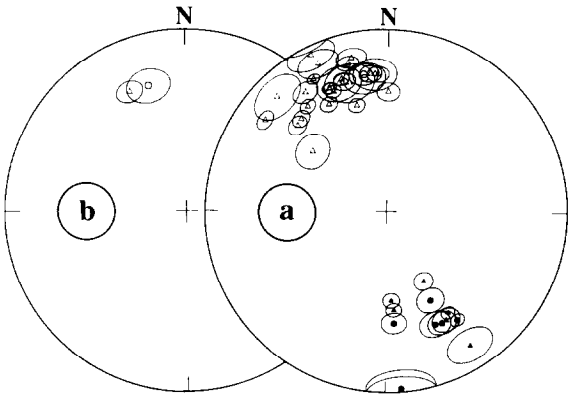


Fig. 6. (a) Site-mean directions with their 95% confidence circles, after tilt corrections. Triangles and circles represent Cretaceous and Cainozoic directions, respectively. Same conventions as Fig. 5c. (b) Mean Cretaceous (△) and Cainozoic (○) directions with their 95% confidence circle.

teristic palaeomagnetic directions calculated for each site (using Fisher's statistic) are shown in Figure 6a and reported in Table 1. Depending on the sites, the two polarities are represented. Also, a fold test performed on two Cretaceous sites (PE116 and PE117) was positive at the 1% significance level (McElhinny, 1973) and more generally there is a significantly better grouping of all the data after tilt corrections. The palaeomagnetic directions do not appear to be related to either the location of the sites or the age of the sampled formations (Fig. 1 relative to declinations). Indeed, a significant scatter is sometimes observed between the data from sites very close to each other for both Cretaceous and Cainozoic volcanic and intrusive directions. This scatter may in principle arise from secular variation, the presence of minor local tectonic disturbances, and/or undetected tilts. However, the secular variation is largely averaged out by the large number of the studied formations and by their good distribution in age. Thus, a mean Cretaceous ($D = -25.1°$; $I = -28.4°$; $\alpha_{95} = 5.3°$) and a mean Cainozoic (D $= -16.5°$; I $= -29°$; $\alpha_{95} = 8.6°$) regional palaeomagnetic direction can be calculated. There is no significant difference between these two directions, although the Cainozoic one has a lower declination of a little less than 10° (Fig. 6b). These directions determine a Cretaceous pole (65.3°N; 180.1°E; $K = 27.3$; $A_{95} = 4.9°$), and a Cainozoic pole (73.2°N; 176.6°E; $K = 43.6$; $A_{95} =$ 7.1°) for the Western Cordillera at the latitude of Lima.

Interpretation of results

To assess the meaning of these results in terms of the development of the Andean Cordillera, the data must be compared to those from SOAM.

The problem of South-American reference poles

A catalog of these poles was first published by Irving and Irving (1982), and recently updated by Beck (1988) who made a critical selection of the available Cretaceous poles according to precise criteria. The good grouping of such selected poles led him to propose a mean palaeomagnetic pole (85.5°N; 253.5°E; $A_{95} = 2.4°$) for SOAM, close to the present one. The author proposed this pole as suitable for practically all the duration of the Cretaceous (from 128 to 75 Ma). Recently, Montes-Lauar and Pacca (1989) and Butler et al. (1989) obtained new Late Cretaceous and Paleocene poles in the southeast of Brazil, and in the south of Chile and Argentina, close to each other, and which determine a mean Late Cretaceous–Paleocene pole of coordinates: 80.7°N; 175.8°E; $A_{95} = 4.7°$. This pole (pole A, Fig. 7b) appreciably differs from the mean South-American pole computed by Beck (pole B, Fig. 7b).

The validity of these different poles may be checked by transferring to South America the palaeopoles from other continents surrounding the Atlantic Ocean, using the kinematics of the opening of this ocean (Klitgord and Schouten, 1986). In this way, the transfer of the North-American and African (or Eurasiatic) polar wander curves of Westphal et al. (1986) leads to South-American Late Cretaceous and Paleocene pseudo-palaeopoles (Fig. 7a) which appear to indicate that SOAM oscillated back and forth during the spreading of the South Atlantic Ocean with an amplitude of about 12° (curves 1 and 2, Fig. 7c). These oscillations, which also appear but in an opposite sense (curve 3, Fig. 7c) when referring to the South-American curve of Irving and Irving (1982), are questioned by Beck (1988). However, taking into account the uncertainties of

TABLE 1

Summary of reliable palaeomagnetic results from Cretaceous and Cainozoic sites of the Lima region (this study)

Sites	n/N	D(°)	I(°)	Dc(°)	Ic(°)	K	α_{95}(°)	Age (Ma)	Lithology
Cretaceous rocks									
PE8899	10/10	315.6	−13.8			20.7	9.7	78–89	Santa Rosa tonalite
PE88100	9/9	354.6	−25.4			30.6	8.4	78–89	Santa Rosa granodiorite
PE88101	6/9	343.2	−27.3			34.0	9.8	78–89	Santa Rosa granodiorite
PE88102	6/6	334.4	−11.7			57.9	7.5	Alb.	Casma lavas
PE88103	8/8	306.3	−17.4			172.0	3.8	Alb.	Casma volcaniclastics
PE88110	8/11	287.4	−31.9	340.5	−24.4	251.4	3.1	Alb.	Pariatambo limestones
PE88116	8/9	320.2	23.4	308.6	−46.4	40.0	7.8	Alb.	Pariatambo limestone
PE88117	12/12	286.0	−22.6	313.5	−32.4	91.0	4.2	Alb.	Pariatambo limestones
PE88123	11/12	350.5	−30.0	355.9	−25.6	101.0	4.2	Alb.	Casma lavas
PE88124	8/9	339.7	−25.5	344.7	−17.4	153.0	4.0	Alb.	Casma lavas
PE88125	7/7	340.5	−18.5	346.4	−14.8	92.0	5.5	Alb.	Casma volcanoclastics
PE88126	8/9	347.0	−21.6	334.4	−26.4	313.0	2.8	Alb.	Casma lavas
PE88127	12/12	335.5	−28.9			168.0	3.1	102	Patap diorite
PE88128	6/9	333.4	−5.2			45.0	8.5	78–89	Santa Rosa tonalite
PE88129	7/9	147.2	13.9			36.0	8.8	78–89	Santa Rosa tonalite
PE88130	8/8	346.9	−28.2	352.7	−25.0	128.0	4.4	Alb.	Casma lavas
PE88131	7/8	338.9	−26.1	338.9	−26.1	36.0	8.8	73	Humaya adamelite
PE88133	10/10	345.8	−30.1	316.7	−31.3	142.0	3.8	Ce.–Tu.	Jumasha limestones
PE88135	9/10	332.6	−48.4	334.5	−26.6	292.0	2.7	Alb.	Chulec limestones
PE88136	12/14	4.15	−17.8	325.0	−22.0	62.0	5.1	Alb.	Pariatambo limestones
PE88138	10/13	132.8	33.0	150.4	32.1	57.0	5.8	Apt.	Pariahuanca limestones
PE88145	11/12	343.7	−39.2	343.7	−39.2	153.0	3.4	Alb.	Casma pillow lavas
PE88146	14/14	338.2	−20.8	330.4	−18.3	221.0	2.5	Alb.	Casma pillow lavas
PE88147	12/12	0.3	−34.6	0.3	−34.6	76.0	4.6	Alb.	Casma lavas
PE88148	8/8	331.1	−33.6			194.0	3.6	Alb.	Casma lavas
PE88149	9/10	322.5	−28.6			196.5	3.3	Alb.	Casma lavas
PE88150	7/7	147.9	34.8			329.5	2.9	81–83	Tiabaya granodiorite
PE88151	8/10	151.3	54.1			140.0	4.2	81–83	Tiabaya granodiorite
PE88152	8/8	175.1	44.5			224.0	3.3	81–83	Tiabaya granodiorite
PE88153	8/8	176.3	49.1			185.5	3.6	81–83	Tiabaya granodiorite
Mean paleomagnetic direction: $n/N = 30/30$; $D = -25.1°$; $I = -28.4°$; $K = 22.9$; $\alpha_{95} = 5.3°$ $R = -15.9° \pm 6°$; $F = -0.9° \pm 7.4°$									
Cainozoic rocks									
PE88104	10/10	152.9	44.4			56.2	5.9	18	Surco granite
PE88105	9/9	175.5	38.0			82.0	5.2	5.4	Surco rhyolite
PE88106	12/12	174.5	2.0			12.5	11.4	18	Surco granite
PE88107	10/15	155.8	32.6			47.0	6.4	15–53	Calipuy volcanoclastics
PE88122	11/11	349.9	−24.8			53.9	5.7	24	Catahuasi granite
PE88132	8/8	350.0	−26.8			37.0	8.1	61–62	Sayan adamelite
PE88140	6/8	341.3	−26.2			164.0	4.4	19–31	Churin granitic stock
PE88142	6/6	146.0	29.2			356.0	3.0	15–53	Calipuy lavas
PE88143	6/8	152.7	31.8			97.0	5.8	61–62	Sayan monzogranite
Mean paleomagnetic direction: $n/N = 9/9$; $D = -16.5°$; $I = -29°$; $K = 29.9$; $\alpha_{95} = 8.6°$ $R = -13.7° \pm 8.1°$; $F = 1.1° \pm 8.1°$									

N = total number of samples; n = number of retained samples; D, I = declination and inclination without tilt correction; Dc, Ic = declination and inclination after tilt correction; K = Fisher's dispersion parameter; α_{95} = 95% confidence circle. Reported Cretaceous and Cainozoic ages are from Giletti and Day (1968), Wilson (1975), Farrar and Noble (1976), Cobbing et al. (1981), Beckinsale et al. (1985), and Mukasa and Tilton (1985). Alb. = Albian; Apt. = Aptian; Ce.-Tu. = Cenomanian–Turonian.

The rotation (R) and flattening (F) parameters are calculated using equations given by Demarest (1983). Reference poles used are given in the text.

the Atlantic Ocean kinematic parameters, the obtained distribution of the pseudo-palaeopoles is close to the mean pole calculated from the new Late Cretaceous–Paleocene poles (pole A, Fig. 7b), and on the contrary is distant from the pole proposed by Beck (pole B, Fig. 7b). Consequently, pole A will be our reference pole for the interpretation of our Cretaceous data. On the basis of this pole, the sampled Cretaceous sites, reduced to a mean geographic location of (11.5°S, 76.5°W) are expected to have a palaeomagnetic declination of about 351° if they had passively followed the movement of the plate without any additional deformation (had the pole proposed by Beck been used instead, the results would not have changed significantly).

For the Cainozoic, because the ages of our sites vary from 62 to 5 Ma, we separated them into four groups depending on their age, in order to compare them to the available SOAM poles. Consequently, sites PE132 and PE143 are referred to the Late Cretaceous–Paleocene pole of Butler et al. (1989), sites PE107, PE140 and PE142 are referred to the pole calculated for 40 Ma in Mitouard et al. (1990), sites PE104, PE106 and PE122 to the 15 Ma pole of Valencio and Creer (1968). Finally the Pliocene site PE105 is referred to the pole of Valencio and Mendia (1974). These poles correspond to poles C, D, E and F of Fig. 7b, respectively.

Evaluation of the amplitude of rotations of the Western Cordillera.

According to the data relative to SOAM, it seems that between the end of the Jurassic and the Present, the rotational displacements with respect to the Earth's axis of this continent have never exceeded 15° (Fig. 7c). Within the uncertainties on the SOAM reference poles, our re-

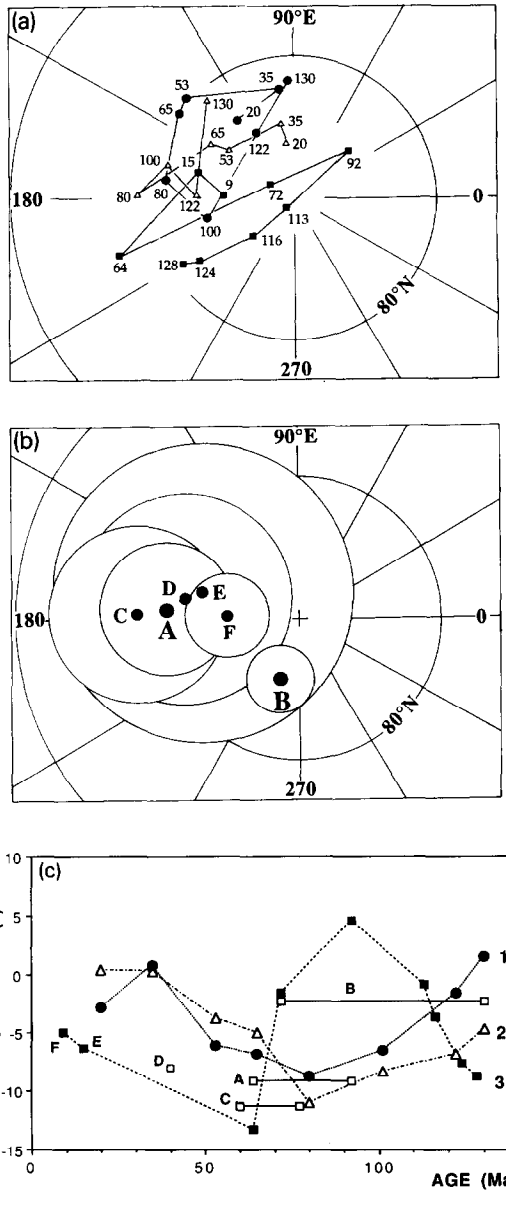


Fig. 7. (a) Early Cretaceous to Present South-American apparent polar wander path (■; Irving and Irving, 1982, 30 Ma window), and North-American (●) and African (△) curves (Westphal et al., 1986) transferred to South America, ages of poles are in Ma. (b) *A* = mean Late Cretaceous–Paleocene pole computed using poles from Montes-Lauar and Pacca (1989) and pole from Butler et al. (1989); *B* = Cretaceous mean pole computed by Beck (1988); *C* = Late Cretaceous–Paleocene pole from Butler et al. (1989); *D* = Eocene pole given in Mitouard et al. (1990); *E* = Miocene pole of Valencio and Creer (1968); *F* = Pliocene pole of Valencio and Mendia (1974). Poles are represented with there 95% confidence circles. (c) Theoretical declinations expected for SOAM between Early Cretaceous to Present in the region of Lima and computed using: *1*, *2* = Westphal's et al. (1986) North-American and synthetic African curves, respectively, transferred to South America; *3* = Irving and Irving's (1982) South-American curve (30 Ma window). Same conventions as for (b).

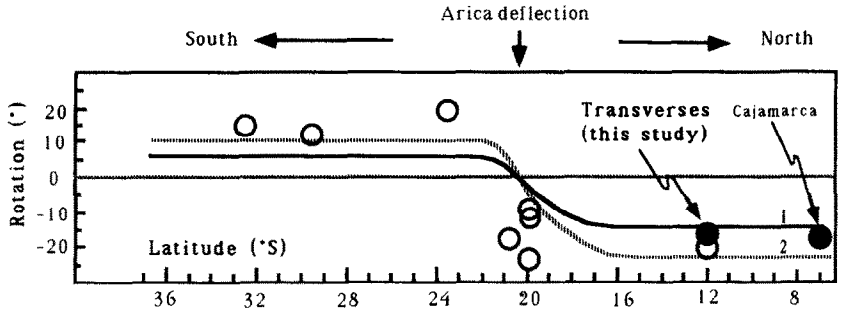


Fig. 8. Predicted values of rotation along the Andean Cordillera according to the two shortening models (1, 2) given in Isacks (1988). The solid circles indicate the amplitudes of rotations inferred from our palaeomagnetic data from the transverses (this study) and the Cajamarca area (Mitouard et al., 1990). Open circles refer to previous data, mainly Mesozoic in age.

sults, obtained from Cretaceous to Late Eocene sites of the Western Cordillera in the region of Lima, show a consistent counterclockwise rotation of about 15–20° without any gradient with respect to the age or to the geographical location of the sites (Table 1). On the other hand, the Miocene and Pliocene reliable sites are not numerous enough to yield any definitive conclusion about the amount of rotation which occurred in the same region during the late Cainozoic. Thus, we prefer only to conclude from this study that the observed counterclockwise rotation is post-Late Eocene in age without any known upper time limit, although Neogene data obtained by Heki et al. (1985a) and Tsunakawa et al. (1987), would give a Late Miocene age for the end of this rotation. The confirmation of this point requires an additional study of a greater number of well-dated Cainozoic sites, particularly of Miocene age.

The consistency between the amplitudes of the Cretaceous to Eocene rotations in the whole studied area, and with the amplitudes of the post-Early Oligocene rotations obtained in the region of Cajamarca (Mitouard et al., 1990) is remarkable. Such a consistency over a distance of about 600 km and from the coast up to the Altiplano, seems more in agreement with a general counterclockwise rotation of the order of 15–20° of the Western Cordillera, than with local in-situ block rotations in a zone of distributed shear. The sense of this rotation about a pole situated northward in the Huancabamba area, agrees with the geological observations which show an increasing amount of shortening going southward from Huancabamba to Arica.

In order to estimate the amount of horizontal displacement related to the rotation, we have introduced the values of the post-Eocene rotations measured both in these three east–west transverses and in the Cajamarca area near 7°S (Mitouard et al., 1990) in the model proposed by Isacks (1988). These data (Fig. 8) seem to be more in agreement with model 1 calculated by the author, and which then predicts a maximum shortening of about 350 km. This is of the same order of magnitude as the amount of shortening (100–250 km) calculated on the basis of gravimetric data (Lyon-Caen et al., 1985), or by the construction of balanced cross-sections (Sheffels, 1988, 1990; Baby et al., 1989). These last data, were obtained in the sub-Andean Ranges of northwest Bolivia, and take into account only part of the shortenings, because the thrusts and overthrusts also develop within the Altiplano and the Eastern Cordillera where probably the major part of the Andean shortenings is consumed (Roeder, 1988; Sempere et al., 1990). One can notice that, had the Western Cordillera behaved as a perfectly rigid elongated lithospheric fragment, the rotation would imply a shortening of about 530 km at the Arica latitude. This value is of course a maximum, since it is not realistic to assimilate the Western Cordillera to a perfectly rigid body at the scale of the Andean Cordillera.

Conclusions

These results, consistent with the ones previously obtained near 7°S (Mitouard et al., 1990), have confirmed the existence of a counterclockwise rotation with an amplitude of 15–20° of the

Western Cordillera during Cainozoic times. The consistency of the rotations measured in the entire Western Cordillera also indicates that the damping of these rotations takes place farther east.

This rotation is in good agreement with the model proposed by Isacks (1988) for the uplift of the Central Andes and appear to be in good agreement with the geological evaluation of the shortenings. However, to unambiguously relate the rotation of the Peruvian margin to the uplift of the Andes, much more palaeomagnetic data are needed, in particular for the Neogene period. Indeed, the precise timing of the rotation is presently not better constrained than post-Late Eocene. A more precise determination of this shortening on the basis of palaeomagnetic data would require a substantial improvement in the definition of the directions of the SOAM reference poles.

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